

PIM solutions for AI SoCs

AI SoC should start adopting processing-in-memory (PIM) solutions to mitigate this bottleneck. PIM is a memory solution that combines logic functions and memory, allowing data processing at lower-level memories while also handling data with PEs with upper-level memory. PIM gets fabricated inside high-bandwidth memory (HBM), which not only brings the best of computing (within the memory) but also data transfer (to and from memory).

So far, Samsung and SK Hynix have developed a PIM memory solution. The data shared shows an 80-85% reduction in power consumption. It is a significant savings, given that AI applications are moving the computing industry towards TOPS—trillions of operations per second. Any solution that can increase the TOPS while improving performance

per watt will be a game changer for the AI SoC domain.

The path forward for AI SoCs

Currently, there are no mass-produced AI SoCs with PIM-based

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features. With the promising solutions from Samsung and SK Hynix (with Micron also re-exploring PIM), there is a strong case for developing AI SoCs using this new memory

architecture. It will not only speed up the processing request of AI applications, but when combined with more-than-Moore solutions like chiplets, it can revolutionise how server-grade data centres get designed, reducing the number of server racks—making data centres more energy efficient.

AI SoCs with PIM will also require several system software-level changes. The applications must manage data processing with PEs and also with PIMs, all concurrently, without introducing a clock cycle penalty.

In summary, a memory-level solution that can speed up the data flow movement and simultaneously lower the number of clock cycles required to process trillions of data points will be a game changer for AI SoC. In this regard, PIM-powered AI SoC is definitely one such solution. **EFY**

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